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CMPUT 365: Introduction to Reinforcement Learning

Worksheet 5: Monte Carlo Methods

Question 1. In Policy Iteration, we used dynamic programming for the policy evaluation step, to compute v_π. Monte Carlo ES is a Generalized Policy Iteration (GPI) algorithm, that does not do a full policy evaluation step, before greedifying. How might you modify Monte Carlo ES, to do (more) complete policy evaluation steps before greedifying?

Loop policy evaluation forever or until you know that the greedy policy will never change. Remove previous estimates before every evaluation step (not required if looping forever but will not change the process). In policy evaluation step, apply the Bellman equation over and over until convergence as opposed to one pass over the state space.

Question 2. Off-policy Monte Carlo prediction allows us to use sample trajectories to estimate the value function for a policy that may be different than the one used to generate the data. Consider the following MDP, with two states B and C, with 1 action in state B and two actions in state C, with $\gamma = 1.0$. In state C both actions transition to the terminating state with $A = 1$ following the blue path to receive a reward $R = 1$ and $A = 2$ following the green path to receive a reward $R = 10$. Assume the target policy π has $\pi(A = 1|C) = 0.9$ and $\pi(A = 2|C) = 0.1$, and that the behaviour policy b has $b(A = 1|C) = 0.25$ and $b(A = 2|C) = 0.75$.

(a) What are the true values v_π?

(b) Imagine you got to execute π in the environment for one episode, and observed the episode trajectory $S_0 = B, A_0 = 1, R_1 = 1, S_1 = C, A_1 = 1, R_2 = 1$. What is the return for B for this episode? Additionally, what are the value estimates V_π, using this one episode with Monte Carlo updates?

(c) But, you do not actually get to execute π; the agent follows the behaviour policy b. Instead, you get one episode when following b, and observed the episode trajectory $S_0 = B, A_0 = 1, R_1 = 1, S_1 = C, A_1 = 2, R_2 = 10$. What is the return for B for this episode? Notice that this is a return for the behaviour policy, and using it with Monte Carlo updates (without importance sampling ratios) would give you value estimates for b.

(d) But, we do not actually want to estimate the values for behaviour b, we want to estimates the values for π. So, we need to use importance sampling ratios for this return. What is the return for B using this episode, but now with importance sampling ratios? Additionally, what is the resulting value estimate for V_π using this return?

(a)
$$\begin{aligned} v_\pi(C) &= \mathbb{E}[R_{t+1} + \gamma v_\pi(\top)|S_t = C] \\ &= \pi(A = 1|C) \times 1 + \pi(A = 2|C) \times 10 \\ &= 0.9 \times 1 + 0.1 \times 10 = 1.9 \\ v_\pi(B) &= \mathbb{E}[R_{t+1} + \gamma v_\pi(C)|S_t = B] \\ &= 1 + v_\pi(C) \\ &= 2.9 \end{aligned}$$

(b) $G_0 = R_1 + \gamma R_2 = 1 + 1 = 2$

$v_\pi(C) = 1$
$v_\pi(B) = 2$

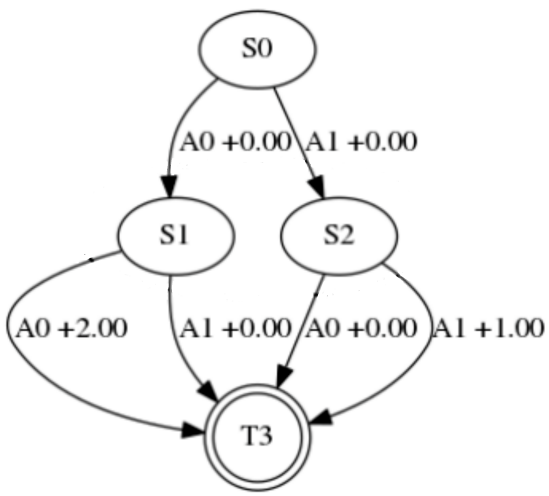
(c) $G_0 = R_1 + \gamma R_2 = 1 + 10 = 11$

(d) $\rho_0 = \dfrac{\pi(A_0 = 1|S_0 = B)}{b(A_0 = 1|S_0 = B)} = 1$

$\rho_1 = \dfrac{\pi(A_1 = 2|S_1 = C)}{b(A_1 = 2|S_1 = C)} = \dfrac{0.1}{0.75} = 0.1333...$

$\rho_0\rho_1 G_0 = 1 \times 0.133 \times 11 \approx 1.47$

Question 3. Consider the three-state MDP below with terminal state T_3 and $\gamma = 1$. Suppose you observe three episodes: $\{S_0, S_1, T_3\}$ with a return of 2, S_0, S_1, T_3 with a return of 2, S_0, S_2, T_3 with a return of 1. What is the (every-visit) Monte-Carlo estimator of the value for each of the states S_0, S_1, S_2? How would the Monte-Carlo estimates change if $r(S_0, A_1, S_2) = +1.00$?



Trajectories:	Returns:
$S_0, A_0, 0, S_1, A_0, 2, T$	$0 + 2 = 2$
$S_0, A_0, 0, S_1, A_0, 2, T$	$0 + 2 = 2$
$S_0, A_1, 0, S_2, A_1, 1, T$	$0 + 1 = 1$

Returns from $S_2 : [1] \rightarrow V(S_2) = avg([1]) = 1$
Returns from $S_1 : [2, 2] \rightarrow V(S_1) = avg([2, 2]) = 2$
Returns from $S_0 : [1, 2, 2] \rightarrow V(S_0) = avg([1, 2, 2]) = 5/3$

Now, if $r(S_0, A_1, S_1) = +1.00$, the last trajectory enumerated above would lead to a return of 2, not 1. This means that the return from S_0 would become $[2, 2, 2] \rightarrow V(S_0) = avg([2, 2, 2]) = 2$. The other estimates would remain unchanged.

Question 4. Consider the three-state MDP from Question 3, but let $r(S_1, A_1, T_3) = +1.00$ and $r(S_2, A_0, T_3) = +2.00$. You observe the following episodes: $\{S_0, S_1, T_3\}$ with a return of 2, $\{S_0, S_1, T_3\}$ with a return of 2, $\{S_0, S_1, T_3\}$ with a return of 1, $\{S_0, S_2, T_3\}$ with a return of 1. What is the (every-visit) Monte-Carlo estimator of the value for each of the states S_0, S_1, S_2?

Trajectories:	Returns:
S_0, A_0, S_1, A_2, T	$0 + 2 = 2$
S_0, A_0, S_1, A_2, T	$0 + 2 = 2$

S_0, A_0, S_1, A_1, T $\quad 0 + 1 = 1$
S_0, A_1, S_2, A_1, T $\quad 0 + 1 = 1$

Returns from $S_2 : [1] \to V(S_2) = avg([1]) = 1$
Returns from $S_1 : [2, 2, 1] \to V(S_1) = avg([2, 2, 1]) = 5/3$
Returns from $S_0 : [2, 2, 1, 1] \to V(S_0) = avg([2, 2, 1, 1]) = 6/4 = 3/2$

Question 5. Suppose you would like to estimate the expected outcome for $3X^2$ where X is the outcome of a fair die. You run an experiment and observe the following sequence of rolls

$$1, 1, 6, 3, 4.$$

How would you use the above data to estimate the expected value $\mathbb{E}[3X^2]$ for a fair die if the above rolls where observed from a loaded die with $P(X = 1) = 1/2$ and $P(X = i) = 1/10$ for $i = 2, 3, 4, 5, 6$?

$$\begin{aligned}
\hat{v} &= 3 \times \left(\frac{1/6}{1/2} \times (1^2 + 1^2) + \frac{1/6}{1/10} \times (6^2 + 3^2 + 4^2) \right) \Big/ 5 \\
&= 3 \times \left(\frac{1}{3} \times 2 + \frac{5}{3} \times 61 \right) \Big/ 5 \\
&= \left(2 + 5 \times 61 \right) \Big/ 5 \\
&= 307/5 = 61.4
\end{aligned}$$

Question 6. Off-policy Monte Carlo prediction allows us to use sample trajectories to estimate the value function for a policy that may be different than the one used to generate the data. Assume the behaviour policy is uniform random: it takes each action with equal probability in each state. The environment has two possible states $\{1, 2\}$ and two possible actions in each state $\{1, 2\}$, with $\gamma = 1.0$. Suppose we collect the following episode, by following the behaviour policy

$$S_0 = 1, A_0 = 2, R_1 = 3, S_1 = 2, A_1 = 1, R_2 = -4,$$

But, we want to evaluate target policy π, which selects action 1 with 0.3 probability in each state. Give an estimate of $V_\pi(S = 1)$ and $V_\pi(S = 2)$, using the episode generated under b.

$$\rho_{0:0} = \frac{0.7}{0.5} = 1.4$$

$$\rho_{0:1} = 1.4 \times \frac{0.3}{0.5} = 0.84$$

$V(1) = \rho \times (3 - 4)$ $\quad$ where $\rho = \dfrac{0.7}{0.5} \times \dfrac{0.3}{0.5}$, so the estimate is -0.84.

$V(2) = \rho \times -4$ $\quad$ where $\rho = \dfrac{0.3}{0.5}$, so the estimate is -2.4.

Question 7. Exercise 5.2 from Sutton & Barto (2018).

Dr. Sutton has asked that the solutions to the textbook exercises not be made electronically available to students. You are encouraged to attend office hours, where the teaching assistants can discuss the exercises with you and show you the answers. Alternatively, if you email Dr. Sutton your answers for an entire chapter, he will likely send you the electronic answer sheet for that chapter.

Question 8. Exercise 5.3 from Sutton & Barto (2018).

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Question 9. Exercise 5.4 from Sutton & Barto (2018).

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Question 10. Exercise 5.5 from Sutton & Barto (2018).

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Question 11. Exercise 5.11 from Sutton & Barto (2018).

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